

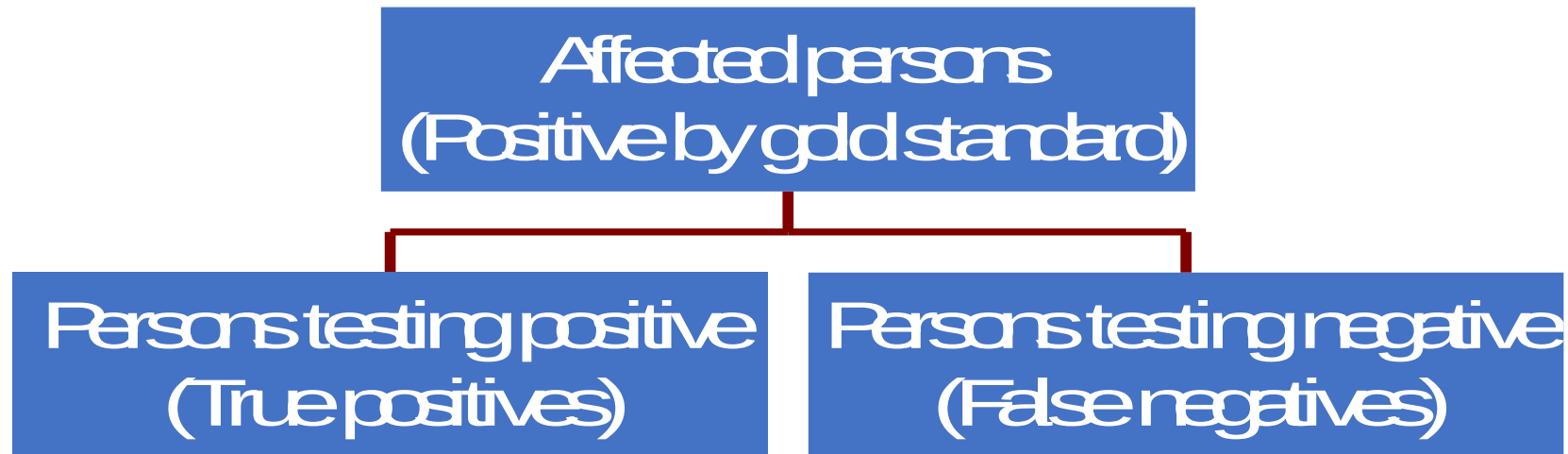
Sensitivity, Specificity, Positive Predictive Value, and Negative Predictive Value

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Sensitivity

The sensitivity of a test is the ability of the test to identify correctly affected individuals

Proportion of persons testing positive among affected individuals



$$\text{Sensitivity} = \text{True positives} / \text{Affected persons}$$

Estimating the sensitivity of a test

- Identify affected individuals with a gold standard
- Obtain a wide panel of samples that are representative of the population of affected individuals
 - Recent and old cases
 - Severe and mild cases
 - Various ages and sexes
- Test the affected individuals
- Estimate the proportion of affected individuals that are positive with the test

Estimating the sensitivity of a urine pregnancy test

- Identify pregnant ladies with a gold standard
 - Using blood test
- Obtain a wide panel of samples that are representative of the population of pregnancy ladies
- Test the ladies with the urine pregnancy test
- Estimate the proportion of ladies that are positive with the urine pregnancy test

Sensitivity of a rapid test for leishmaniasis

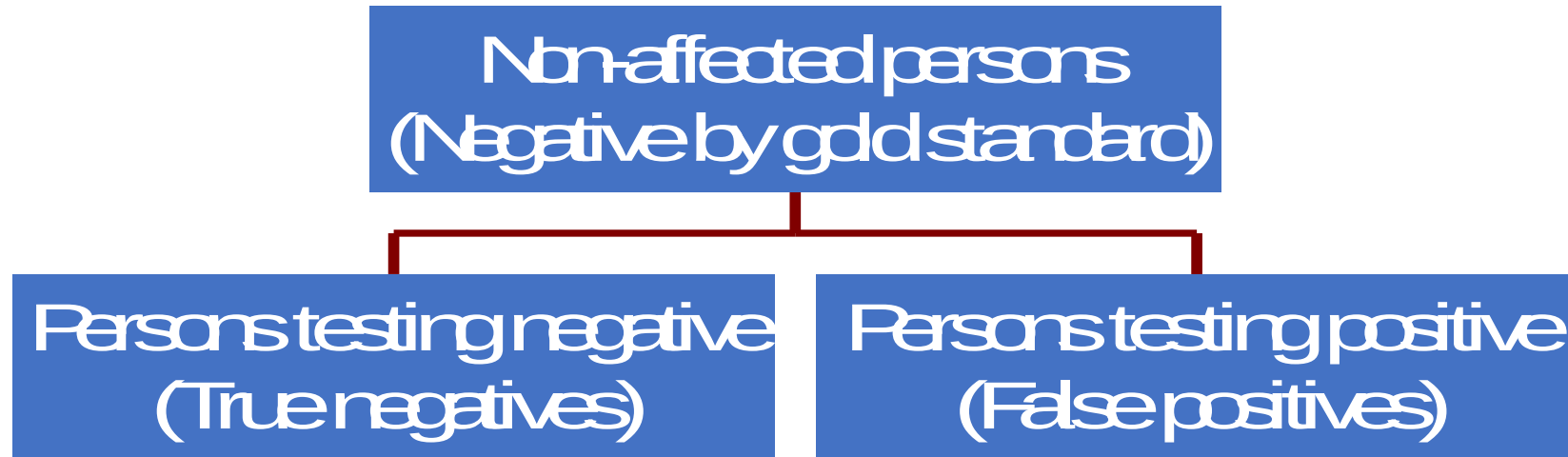
		Pregnant ladies
Urine test	True positive	148
	False negative	2
		150

Sensitivity = $148 / (150) = 98\%$
95% confidence interval: 94%-99%

Specificity

The specificity of a test is the ability of the test to identify correctly non-affected individuals

Proportion of person testing negative among non affected individuals



$$\text{Specificity} = \text{True negatives} / \text{Non-affected persons}$$

Estimating the specificity of a test

- Identify non affected individuals
 - Negative with a gold standard
 - Unlikely to be infected
- Obtain a wide panel of samples that are representative of the population of non-affected individuals
 - Diverse unaffected population: Difficult to find. Ideally, those individuals that would need to be tested but not infected
- Test the non-affected individuals
- Estimate the proportion of non-affected individuals that are negative with the test

Estimating the specificity of a urine pregnancy test

- Identify ladies who are not pregnant with a gold standard
 - Using blood test
- Obtain a wide panel of samples that are representative of the population of pregnancy ladies
- Test the ladies with the urine pregnancy test
- Estimate the proportion of ladies that are negative with the urine pregnancy test

Specificity of a rapid test for leishmaniasis

		Individuals who are not pregnant
Test	False positive	12
	True negative	188
		200

Specificity = $188 / 200 = 94\%$
95% confidence interval: 90%-96%

How is the test doing in a real population?

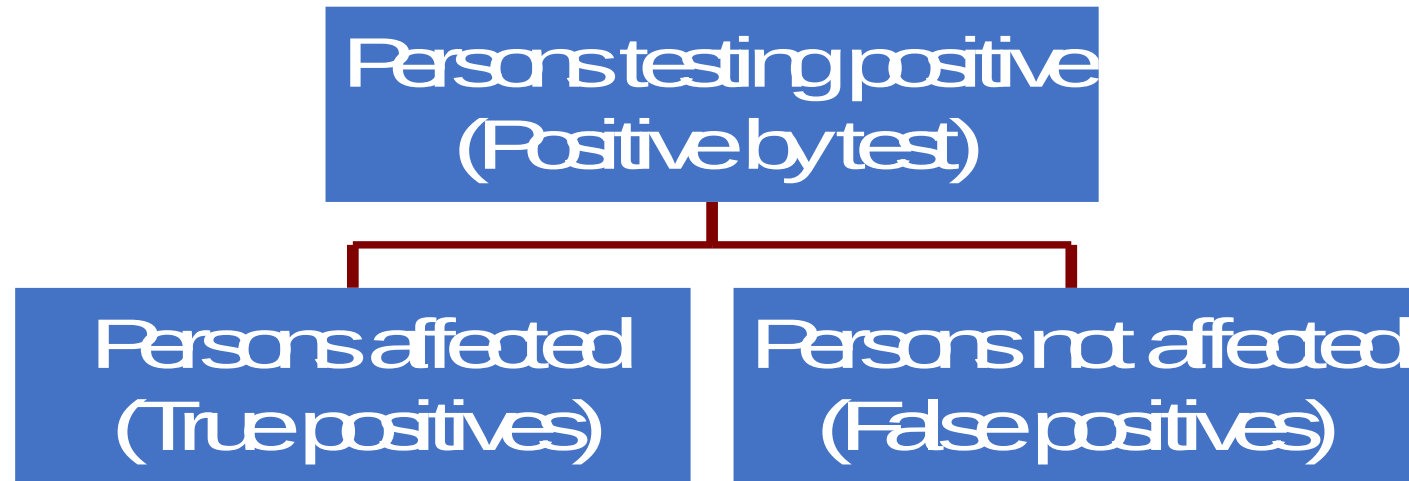
- The test is now used in a real population
- This population is made of
 - Affected individuals
 - Non-affected individuals
- The proportion of affected individuals is the prevalence

		Status of persons		
		Affected	Non-affected	
Test	Positive	True +	False +	A+B
	Negative	False -	True -	C+D
		A + C	B+D	A+C+B+D

Predictive value of a positive test

The predictive value of a positive test is the probability that an individual testing positive is truly affected

Proportion of affected persons among those testing positive



Predictive value of a positive test =
True positives / Persons testing positive

Predictive value of a positive test

		Status of persons		
		Affected	Non-affected	
Test	Positive	A	B	A+B
	Negative	C	D	C+D
		A + C	B+D	A+C+B+D

$$\text{PVP} = A / (A+B)$$

What factors influence the predictive value positive of a test?

- Sensitivity?

- ✓YES: To some extend.

- Specificity?

- ✓YES: The more the test is specific, the more it will be negative for non affected persons. Thus, when the test is positive, it is probably truly positive (All non affected were correctly identified as testing negative).

- Prevalence of the disease?

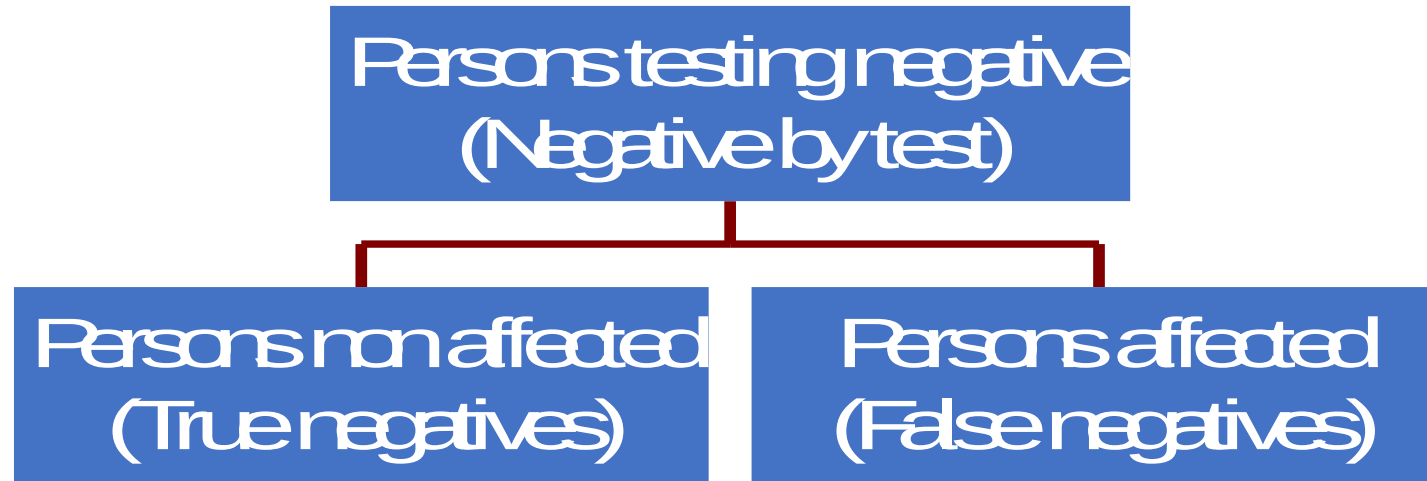
- ✓YES: Low prevalence: The test will pick up more false positives

- ✓YES: High prevalence: The test will pick up more true positives

Predictive value of a negative test

The predictive value of a negative test is the probability that an individual testing negative is truly non-affected

Proportion of non-affected persons among those testing negative



Predictive value of a negative test =
True negatives / Persons testing negative

Predictive value of a negative test

		Status of persons		
		Affected	Non-affected	
Test	Positive	A	B	A+B
	Negative	C	D	C+D
		A + C	B+D	A+C+B+D

$$\text{PVN} = D / (C+D)$$

What factors influence the predictive value negative of a test?

- Sensitivity?
 - ✓ YES: The more the test is sensitive, the more it captures affected persons. Thus, when the test is negative, it is probably truly negative (all affected were captured among the positive).
- Specificity?
 - ✓ YES: But to a lesser extend.
- Prevalence of the disease?
 - ✓ YES: Low prevalence: The test will pick up more true negatives
 - ✓ YES: High prevalence: The test will pick up more false negatives